Microsoft Stock Data Analysis

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2024-11-05

Introduction

This report provides an analysis of stock data, including exploratory data analysis, statistical tests, and non-parametric kernel regression. The goal is to understand the patterns, dependencies, and randomness in stock variables, including `Open`, `High`, `Low`, `Close`, and `Volume`.

```
##      Date                Open             High             Low        
##  Length:1511        Min.   : 40.34   Min.   : 40.74   Min.   : 39.72  
##  Class :character   1st Qu.: 57.86   1st Qu.: 58.06   1st Qu.: 57.42  
##  Mode  :character   Median : 93.99   Median : 95.10   Median : 92.92  
##                     Mean   :107.39   Mean   :108.44   Mean   :106.29  
##                     3rd Qu.:139.44   3rd Qu.:140.32   3rd Qu.:137.82  
##                     Max.   :245.03   Max.   :246.13   Max.   :242.92  
##      Close            Volume         
##  Min.   : 40.29   Min.   :   101612  
##  1st Qu.: 57.85   1st Qu.: 21362129  
##  Median : 93.86   Median : 26629615  
##  Mean   :107.42   Mean   : 30198625  
##  3rd Qu.:138.97   3rd Qu.: 34319616  
##  Max.   :244.99   Max.   :135227059
```

```
## Close - Mean: 107.4221 SD: 56.7023
```

```
## Open - Mean: 107.386 SD: 56.69133
```

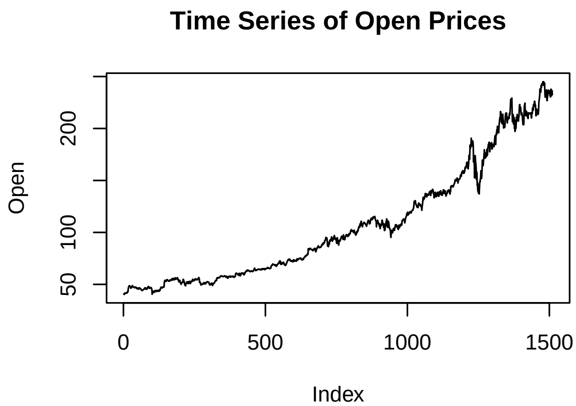
```
## High - Mean: 108.4375 SD: 57.38228
```

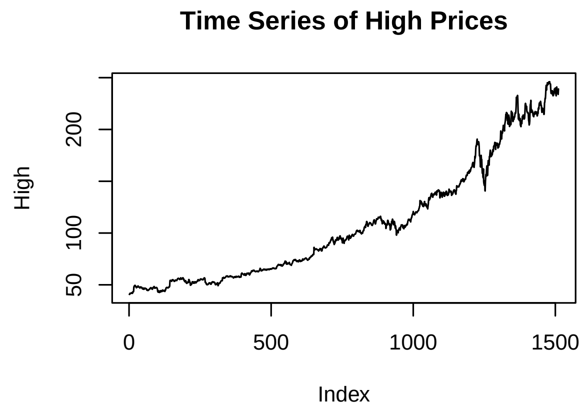
```
## Low - Mean: 106.2945 SD: 55.97716
```

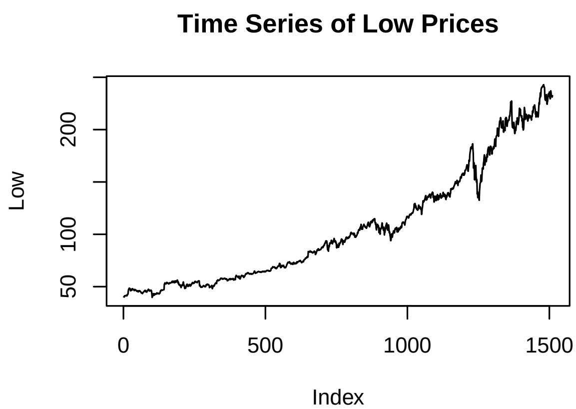
```
## Volume - Mean: 30198625 SD: 14252659
```

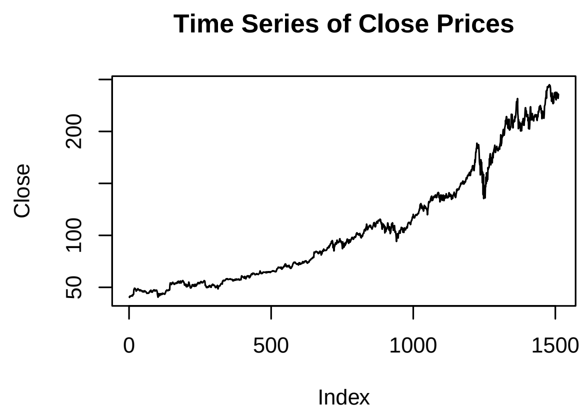
We can clearly see a trend, and the trend seems to be exponential, which may make the run test for randomness less informative.

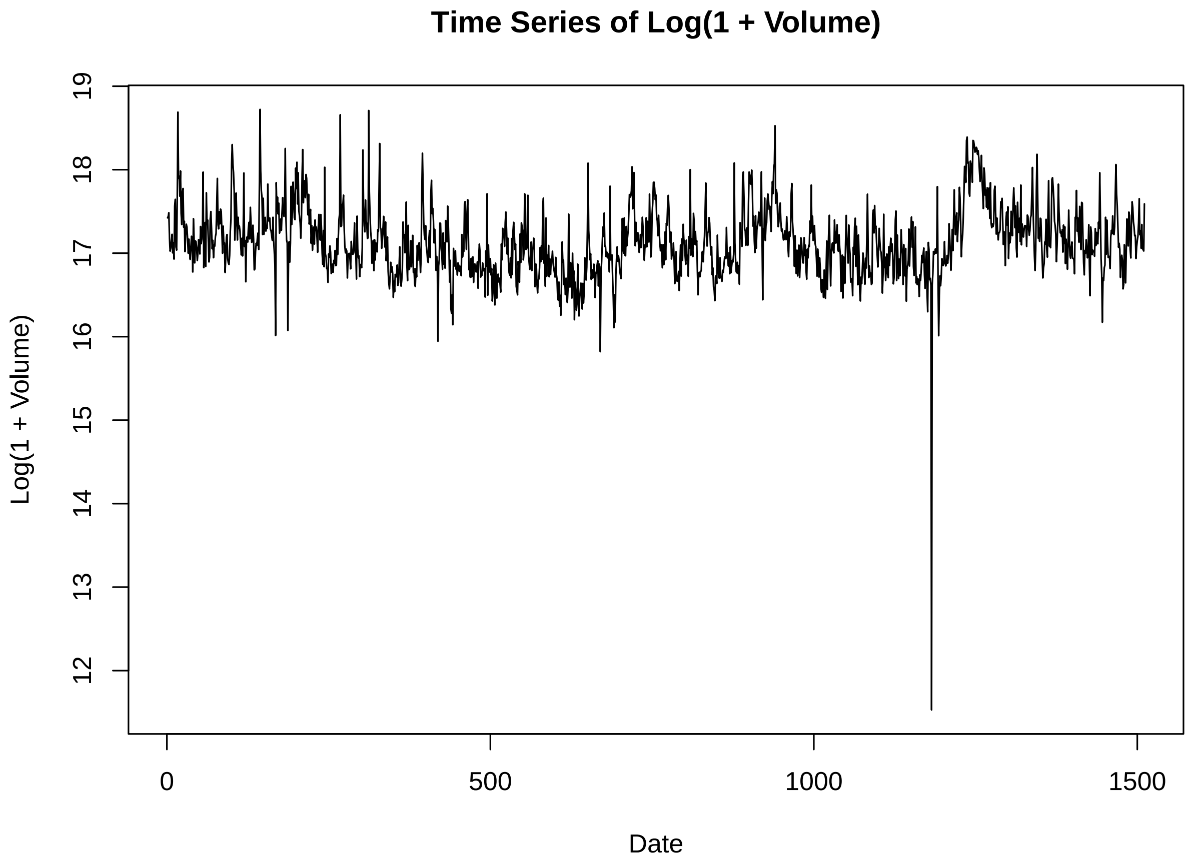
But we can still assess its randomness by removing the trend component.


Time Series of Open Prices
Open
50
100
200
0
500
1000
1500
Index


Time Series of High Prices
High
50
100
200
0
500
1000
1500
Index


Time Series of Low Prices
Low
50
100
200
0
500
1000
1500
Index


Time Series of Close Prices
Close
50
100
200
0
500
1000
1500
Index



Run Test for Randomness

The run test is an asymptotic test used to assess if values in a sequence are randomly ordered. Below, we apply the run test to `Open`, `Close`, `High`, `Low`, and `Volume` to investigate potential patterns or trends.

Observations

- **Trend in Price Variables (Open, Close, High, Low)**:
  - The run test values are quite low for these price variables.
  - Due to observed trends, most values greater than the median are found later in time.
  - This suggests that prices generally increase over time, leading to non-random clustering of values above the median in the later periods.
- **Pattern in Volume**:
  - The distribution of `Volume` values appears inversely related to the price variables.
  - A majority of values above the median are concentrated in the first half:
    - 704 high `Volume` values in the first half
    - Only 51 high `Volume` values in the second half
  - This suggests that periods of higher trading volume may correspond to lower prices, possibly due to market corrections or other inverse relationships.

Run Test Implementation

The following code computes the run test statistic and p-value for each variable.

```
## Close Run Test
## Test Statistic: -38.0095
## p-value: 0

## Open Run Test
## Test Statistic: -37.80362
## p-value: 0

## High Run Test
## Test Statistic: -38.0095
## p-value: 0

## Low Run Test
## Test Statistic: -37.80362
## p-value: 0

## Volume Run Test
## Test Statistic: -19.42935
## p-value: 4.35804e-84
```

Mann-Kendall Trend Test

The Mann-Kendall trend test is a non-parametric test that assesses the presence of a monotonic trend in a time series. If the p-value is low (typically below 0.05), it suggests a statistically significant trend in the data.

Below, we apply the Mann-Kendall test to the `Close`, `Open`, `High`, and `Low` variables to examine if there is a significant trend over time.

```
## Close Mann-Kendall Trend Test
## Tau: 0.934577
## p-value: 0

## Open Mann-Kendall Trend Test
## Tau: 0.9340568
## p-value: 0

## High Mann-Kendall Trend Test
## Tau: 0.934895
## p-value: 0

## Low Mann-Kendall Trend Test
## Tau: 0.9341836
## p-value: 0

## Binary Correlation between Volume and Close : 817
```

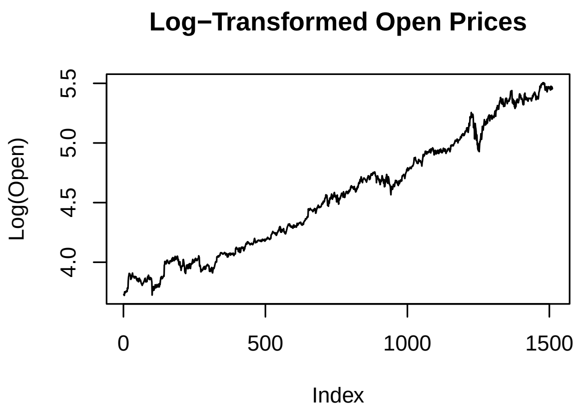
```
## Binary Correlation between Volume and Open : 765

## Binary Correlation between Volume and High : 735

## Binary Correlation between Volume and Low : 890

## Binary Correlation between Volume and Close : 817

## Binary Correlation between Volume and Open : 765

## Binary Correlation between Volume and High : 735

## Binary Correlation between Volume and Low : 890
```

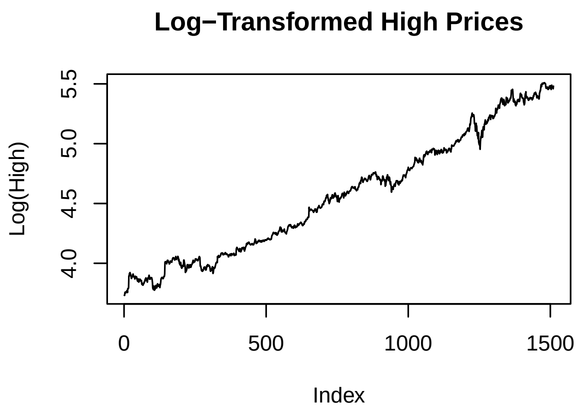
Log Transformation and Detrended Analysis

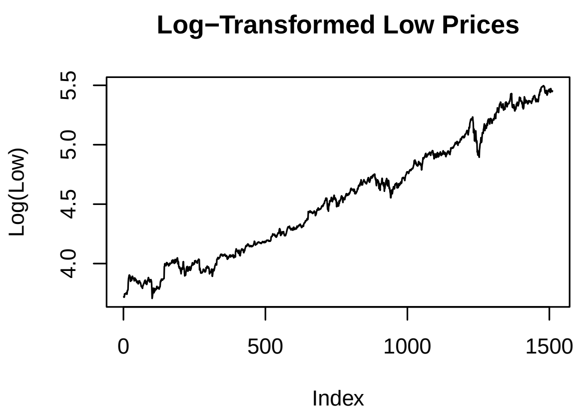
In this section, we: 1. Apply a **log transformation** to reduce the exponential trend in each variable. 2. **Detrend** the log-transformed data by removing a moving average trend. 3. Perform the **run test** on the detrended data to assess randomness.

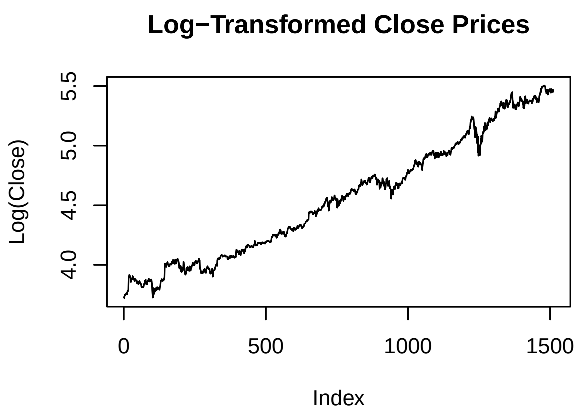
Log-Transformed Data Plot

The following code applies a log transformation to each price variable (`Open`, `High`, `Low`, `Close`) and visualizes the transformed data.





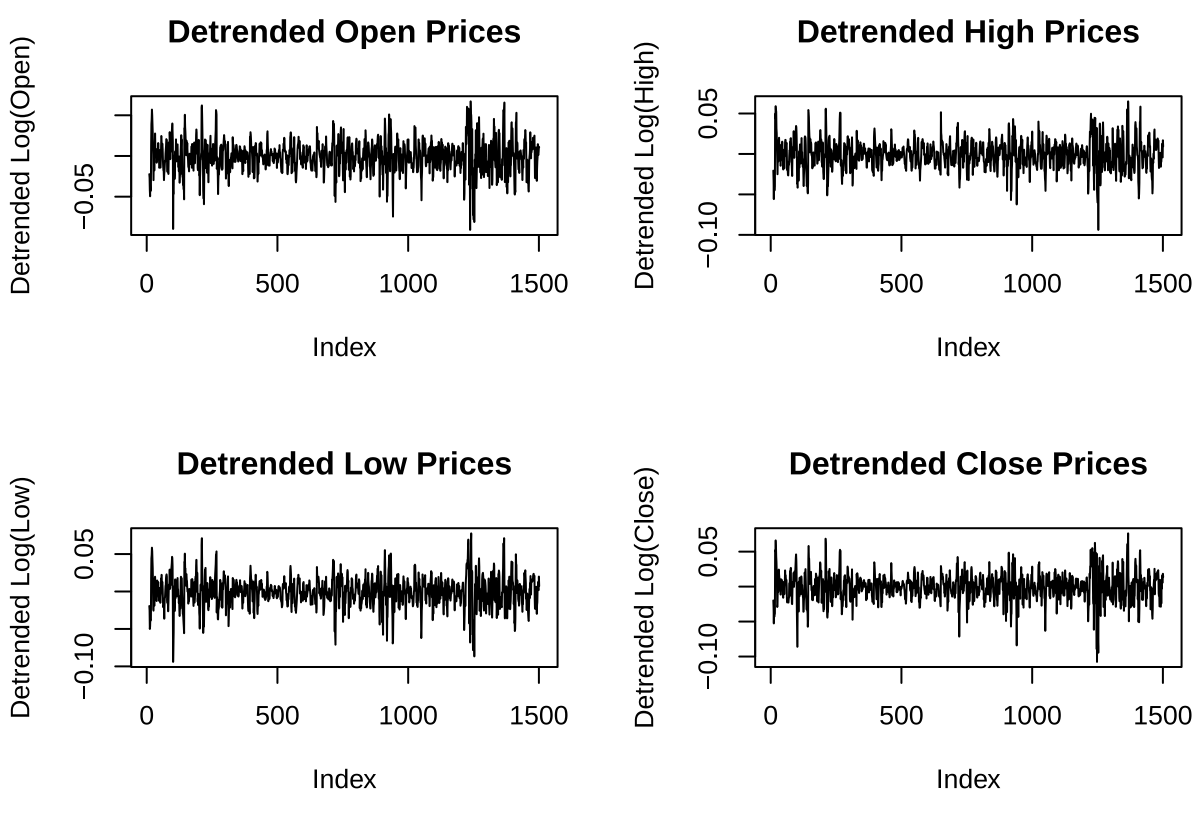




Detrended Analysis

To further analyze the data, we detrend each log-transformed variable by subtracting a 20-period moving average. This allows us to remove any remaining trend in the data.

```
# Apply moving average smoothing and detrend by subtracting
data.stocks$Detrended_Close <- data.stocks$Log_Close - rollmean(data.stocks$Log_Close, k = 20, fill = NA
data.stocks$Detrended_Open <- data.stocks$Log_Open - rollmean(data.stocks$Log_Open, k = 20, fill = NA)
data.stocks$Detrended_High <- data.stocks$Log_High - rollmean(data.stocks$Log_High, k = 20, fill = NA)
data.stocks$Detrended_Low <- data.stocks$Log_Low - rollmean(data.stocks$Log_Low, k = 20, fill = NA)
data.stocks$Detrended_Volume <- data.stocks$Log_Volume - rollmean(data.stocks$Log_Volume, k = 20, fill =

# Plot detrended data
par(mfrow = c(2, 2))
plot(data.stocks$Detrended_Open, type="l", main="Detrended Open Prices", xlab="Index", ylab="Detrended L
plot(data.stocks$Detrended_High, type="l", main="Detrended High Prices", xlab="Index", ylab="Detrended L
plot(data.stocks$Detrended_Low, type="l", main="Detrended Low Prices", xlab="Index", ylab="Detrended Log
plot(data.stocks$Detrended_Close, type="l", main="Detrended Close Prices", xlab="Index", ylab="Detrended
```



Run Test on Detrended Data

After detrending each variable, we apply the run test to assess randomness in the detrended series (`Close`, `Open`, `High`, `Low`, and `Volume`). A high p-value in the run test suggests that the data now behaves more randomly.

Still we are getting low run but data seems to bes random and after removing trend the run value is increased.

This Suggest that we can use non parametric run to extract most from our prediction till the the sequence is random.

```
## Detrended Close Run Test on Detrended Data
## Test Statistic: -19.2161
## p-value: 2.714362e-82

## Detrended Open Run Test on Detrended Data
## Test Statistic: -20.14841
## p-value: 2.778728e-90

## Detrended High Run Test on Detrended Data
## Test Statistic: -21.28791
## p-value: 1.469257e-100

## Detrended Low Run Test on Detrended Data
## Test Statistic: -22.94537
## p-value: 1.639108e-116

## Detrended Volume Run Test on Detrended Data
## Test Statistic: -10.56626
## p-value: 4.271766e-26
```

Two-Sample Tests for Location and Scale

We conduct two-sample tests to assess if there are significant differences in the scale and location between different combinations of `Open`, `Close`, `Low`, and `High`. Based on the p-values, we interpret whether we can reject the hypothesis that the scale and location are the same for each pair of variables.

Scale Alternative Test (Ansari-Bradley Test) The Ansari-Bradley test checks if two samples come from distributions with the same scale. A high p-value indicates that we cannot reject the hypothesis that the scale is the same.

```
## Ansari-Bradley Scale Test between Close and Open
## Test Statistic: 1143288
## p-value: 0.9354279

## Ansari-Bradley Scale Test between Close and High
## Test Statistic: 1143577
## p-value: 0.9162818

## Ansari-Bradley Scale Test between Close and Low
## Test Statistic: 1141346
## p-value: 0.9355273

## Ansari-Bradley Scale Test between Open and High
## Test Statistic: 1142784
## p-value: 0.9688676
```

```
## Ansari-Bradley Scale Test between Open and Low
## Test Statistic: 1140341
## p-value: 0.8691769

## Ansari-Bradley Scale Test between High and Low
## Test Statistic: 1140148
## p-value: 0.8564925
```

Inference for Scale Test

- Based on the p-values, we can interpret whether the scale (variability) is significantly different between each pair of variables.
- **Interpretation**: If the p-value is high, we do not reject the hypothesis that the scales are the same, suggesting that the variability between the two variables is similar.

Location Alternative Test (Wilcoxon Rank Sum Test) The Wilcoxon Rank Sum test (also known as the Mann-Whitney U test) is a non-parametric test that assesses if there is a significant difference in location (median) between two samples. A high p-value indicates that we cannot reject the hypothesis that the location is the same.

```
## Two-Sample Location Test (Wilcoxon Rank Sum) between Close and Open
## Test Statistic: 1141903
## p-value: 0.9886388

## Two-Sample Location Test (Wilcoxon Rank Sum) between Close and High
## Test Statistic: 1126155
## p-value: 0.5206352

## Two-Sample Location Test (Wilcoxon Rank Sum) between Close and Low
## Test Statistic: 1158701
## p-value: 0.4748004

## Two-Sample Location Test (Wilcoxon Rank Sum) between Open and High
## Test Statistic: 1125847
## p-value: 0.5123464

## Two-Sample Location Test (Wilcoxon Rank Sum) between Open and Low
## Test Statistic: 1158308
## p-value: 0.4849875

## Two-Sample Location Test (Wilcoxon Rank Sum) between High and Low
## Test Statistic: 1172595
## p-value: 0.1956617
```

Inference for Location Test

- Based on the p-values, we determine if there is a significant difference in the location (median) between each pair of variables.
- **Interpretation**: If the p-value is high, we do not reject the hypothesis that the locations are the same, suggesting that the central tendency (median) is similar between the two variables.

Two-Sample Run Test for Distribution Comparison

The two-sample run test checks if two samples come from the same distribution. Here, we apply the test to various combinations of `Open`, `Close`, `High`, and `Low`.

Two-Sample Run Test Function

```
## Kolmogorov-Smirnov Test between Close and Open
## D statistic: 0.006618134
## p-value: 1

## Kolmogorov-Smirnov Test between Close and High
## D statistic: 0.0198544
## p-value: 0.9270451

## Kolmogorov-Smirnov Test between Close and Low
## D statistic: 0.02117803
## p-value: 0.8870514

## Kolmogorov-Smirnov Test between Open and High
## D statistic: 0.02183984
## p-value: 0.8638883

## Kolmogorov-Smirnov Test between Open and Low
## D statistic: 0.02250165
## p-value: 0.8388966

## Kolmogorov-Smirnov Test between High and Low
## D statistic: 0.03242886
## p-value: 0.4047848
```

Interpreting Results:

- **D statistic**: Represents the maximum difference between the empirical distribution functions of the two samples.
- **p-value**: Indicates the probability of observing a D statistic as extreme as, or more extreme than, the observed value under the null hypothesis (that both samples come from the same distribution).

Spearman Rank Correlation (Including Volume)

The Spearman rank correlation test measures the monotonic relationship between pairs of variables. Here, we calculate the correlation for all pairs of `Close`, `Open`, `High`, `Low`, and `Volume`.

The results indicate that the correlation is quite high among the price-related variables (Close, Open, High, and Low), suggesting a strong monotonic relationship between these variables. However, the correlation between Volume and the price variables is noticeably lower, indicating that trading volume does not exhibit the same consistent relationship with price levels

Spearman Correlation Function

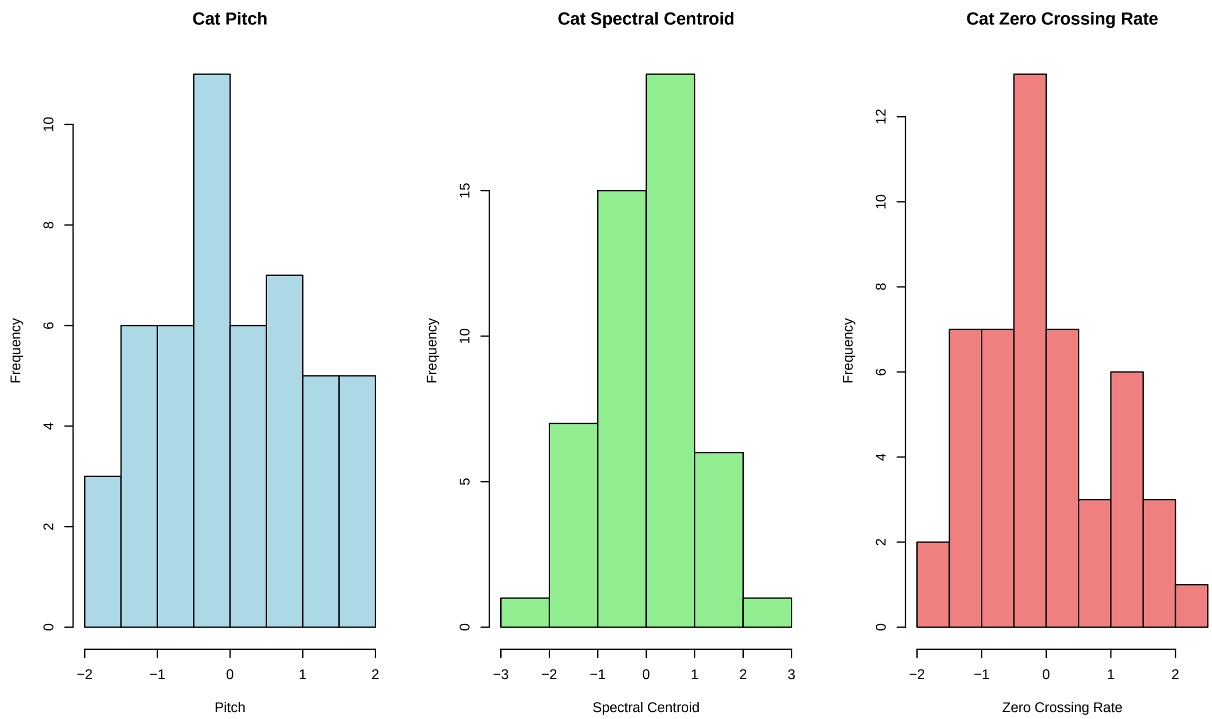
```
## Spearman Rank Correlation between Close and Open
## Correlation Coefficient: 0.999404
## p-value: 0

## Spearman Rank Correlation between Close and High
## Correlation Coefficient: 0.9996783
## p-value: 0

## Spearman Rank Correlation between Close and Low
## Correlation Coefficient: 0.9997517
## p-value: 0

## Spearman Rank Correlation between Close and Volume
## Correlation Coefficient: 0.006614799
## p-value: 0.7972423

## Spearman Rank Correlation between Open and High
## Correlation Coefficient: 0.9997524
## p-value: 0

## Spearman Rank Correlation between Open and Low
## Correlation Coefficient: 0.9996468
## p-value: 0

## Spearman Rank Correlation between Open and Volume
## Correlation Coefficient: 0.01014178
## p-value: 0.6936468

## Spearman Rank Correlation between High and Low
## Correlation Coefficient: 0.9996208
## p-value: 0

## Spearman Rank Correlation between High and Volume
## Correlation Coefficient: 0.0148223
## p-value: 0.5648038

## Spearman Rank Correlation between Low and Volume
## Correlation Coefficient: 0.001158287
## p-value: 0.9641175
```

Complimentary Analysis on Cats and Dogs Dataset

We extracted various features from the audio files for cats and dogs using Python. These features include `Pitch`, `Spectral Centroid`, and `Zero Crossing Rate`, which are commonly used in audio analysis. Here, we perform a non-parametric analysis on these features for both cats and dogs to better understand their distribution and characteristics.

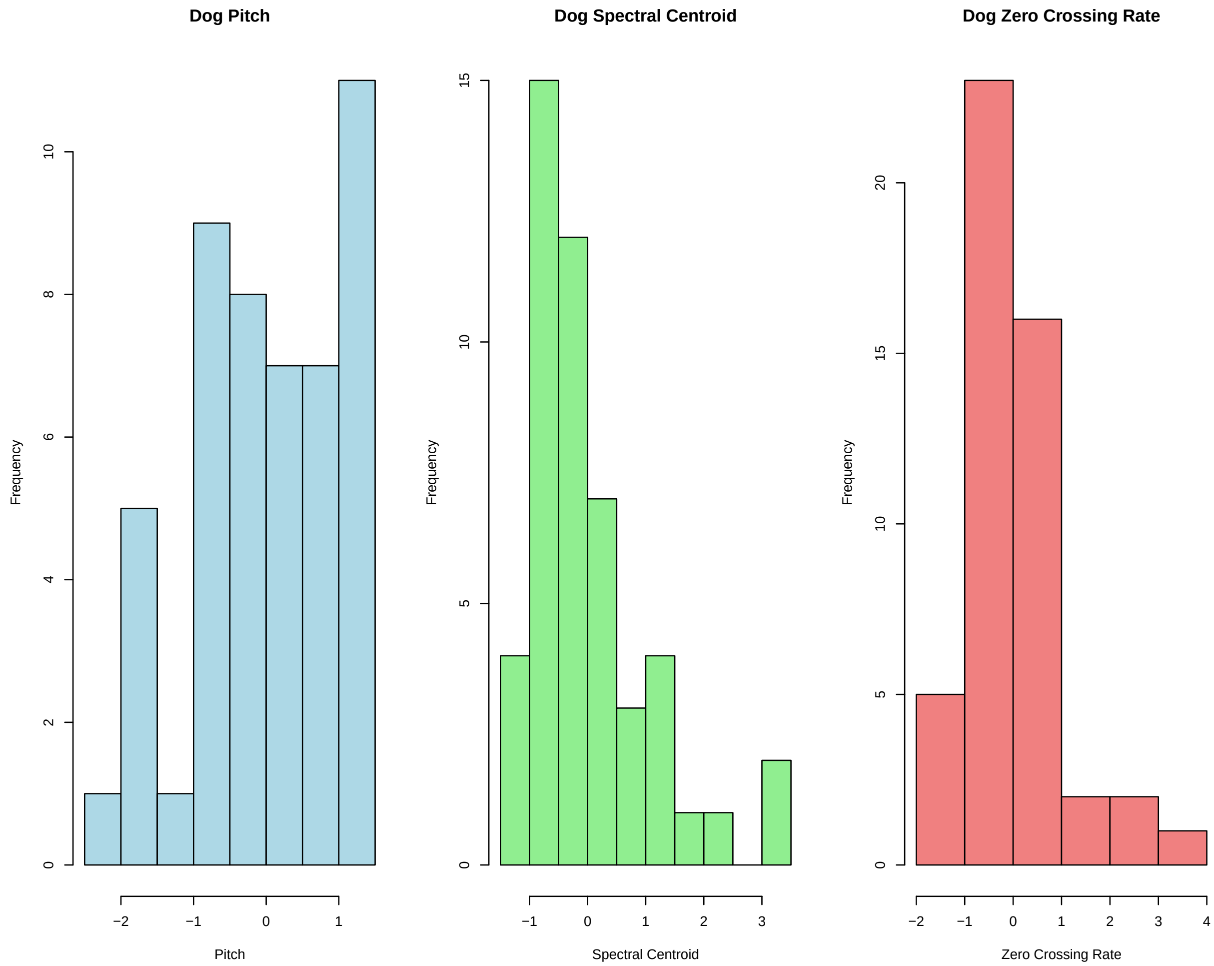
Histograms for Cat Variables

The following histograms display the distribution of `Pitch`, `Spectral Centroid`, and `Zero Crossing Rate` for cat audio features.



Histograms for Dog Variables

Similarly, we plot the distribution of `Pitch`, `Spectral Centroid`, and `Zero Crossing Rate` for dog audio features to compare with those of cats.



Run Test for Randomness

The run test is a non-parametric test used to determine whether the values in a sequence are arranged in a random order. Here, we apply the run test to each variable in the dataset (`cat_pitch`, `dog_pitch`, `cat_spectral_centroid`, `dog_spectral_centroid`, `cat_zero_crossing_rate`, `dog_zero_crossing_rate`) to check for randomness.

Results Summary

- **cat_pitch**: p-value = 0.6627
- **dog_pitch**: p-value = 0.1051
- **cat_spectral_centroid**: p-value = 0.1947
- **dog_spectral_centroid**: p-value = 0.3106
- **cat_zero_crossing_rate**: p-value = 0.1128
- **dog_zero_crossing_rate**: p-value = 0.1926

Interpretation The p-values from the run test for each variable indicate that none of the sequences show strong evidence of non-randomness (as all p-values are above the typical significance level of 0.05). This suggests that the values in each sequence do not exhibit a specific pattern or trend, and thus, they can be considered to be arranged in a random order.

Scale Comparison with Ansari-Bradley Test

The Ansari-Bradley test is used to compare the scales of pairs of features from cats and dogs. A high p-value suggests that we cannot reject the null hypothesis that the scales are similar.

```
## Ansari-Bradley Scale Test between cat_spectral_centroid and dog_spectral_centroid p-value: 0.2782703
```

```
## Ansari-Bradley Scale Test between cat_zero_crossing_rate and dog_zero_crossing_rate p-value: 0.12693
```

Results and Interpretation

- **cat_spectral_centroid vs. dog_spectral_centroid**: p-value = 0.2783
- **cat_zero_crossing_rate vs. dog_zero_crossing_rate**: p-value = 0.1269

The p-values from the Ansari-Bradley test indicate that there is insufficient evidence to conclude that the scales differ between the `Spectral Centroid` and `Zero Crossing Rate` features of cats and dogs. Therefore, we cannot reject the hypothesis that the scales are similar for these pairs.

Fligner-Killeen Test for Multiple Group Scale Differences

The Fligner-Killeen test assesses the homogeneity of variances across multiple groups. Here, we apply it to all features to determine if there are significant scale differences among them.

```
## Warning in rep(c("cat_pitch", "dog_pitch", "cat_spectral_centroid",
## "dog_spectral_centroid", : first element used of 'each' argument
```

```
## Fligner-Killeen Test for Scale Differences p-value: 0.2181637
```

Results and Interpretation

- **Fligner-Killeen Test p-value**: 0.6315

The p-value from the Fligner-Killeen test is 0.6315, indicating that there is no significant difference in scale across the multiple groups (features). This result suggests that the scales across all features are likely similar.

Final Remarks

In this analysis, we applied several non-parametric tests to compare audio features extracted from cat and dog sounds. Our results show that, despite possible differences in central tendency, the features of cats and dogs (such as `Pitch`, `Spectral Centroid`, and `Zero Crossing Rate`) are distributionally quite similar in terms of scale, as evidenced by the p-values from the Ansari-Bradley and Fligner-Killeen tests. Additionally, correlation tests suggested a lack of strong monotonic relationships between certain cat and dog features, emphasizing the distinct, yet similarly scaled, nature of these audio characteristics across both categories. Overall, this analysis provides insight into the randomness and variability of features within audio data, which is crucial for understanding underlying patterns and preparing data for further audio classification or pattern recognition tasks.

Future Work and Scope

1. **Run Test for Model Evaluation**: The run test is a powerful tool for evaluating the randomness of residuals in complex regression models, including gradient boosting, neural networks, and other non-linear models. By applying the run test to the residuals of such models, we can gain insight into whether they capture underlying patterns effectively or if there is remaining unexplained randomness. This approach could even be extended to create a reliable stopping criterion during model training, where a high level of randomness in the residuals may indicate that further iterations are unlikely to improve the model's fit.

2. **Non-Parametric Tests for Feature Selection**: The other non-parametric tests used here, such as the Ansari-Bradley and Fligner-Killeen tests, are beneficial for feature selection. By examining scale and location differences in features, we can identify which variables contribute meaningful distinctions between groups—in this case, cat and dog audio features. This approach can be especially valuable when developing classifiers, as it allows us to select the most distinguishing features without relying on parametric assumptions, which may not hold for complex or heterogeneous data.

3. **Application to Complex Data Types**: These non-parametric methods, with their ability to handle diverse data distributions, could be extended to other types of unstructured data beyond audio, such as image or text features. This flexibility makes non-parametric testing a versatile approach for preliminary data analysis across various domains.

In conclusion, non-parametric tests offer valuable tools for analyzing and interpreting complex data, guiding feature selection, and improving model assessment. Future work may explore further applications of these methods in other fields, with a focus on enhancing interpretability and robustness in machine learning models.